A Middle-Way Equivalence Recasting of the ABC Conjecture

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Abstract

This short note presents a Middle-Way Equivalence (MW-equivalence), denoted $\overset{\text{mw}}{=}$, inspired by the DIFW/EMT framework of dynamic coherence. We show that when the classical equality symbol "="—a static identity relation—is replaced by $\overset{\text{mw}}{=}$, a notion of *dynamic structural convergence*, the classical ABC conjecture admits a direct and structurally stable formulation. This provides a minimal-resolution analogue of Mochizuki's IUT approach without the introduction of new mathematical universes. Instead, it resolves the additive–multiplicative tension by correcting the underlying equivalence operator.

1 Introduction

The classical ABC conjecture concerns integer triples (a, b, c) satisfying $a + b = c$ with $\gcd(a, b) = 1$, and predicts that c cannot be too large relative to the radical $\mathrm{rad}(abc)$. Despite enormous interest, the conjecture remains among the most difficult open problems in number theory.

Recent work on DIFW (Dynamic Indeterminate Framework for Wisdom) and its verification test EMT (Emergent Middle Test) suggests that a large class of mathematical paradoxes, including Gödel incompleteness and P vs. NP, arise from what has been termed a *Non-Middle Fallacy*: the application of a static identity operator to a fundamentally dynamic system.

In this note we apply this viewpoint to ABC. We argue that the classical equality $a + b = c$ conflates two distinct "universes":

- an *additive universe*, where complexity behaves linearly;
- a *multiplicative universe*, where complexity expands via prime factorization.

The equality sign enforces a strict identity between these universes, creating the well-known tension at the heart of ABC.

Our proposal is to replace this static identity with the Middle-Way Equivalence $\overset{\text{mw}}{=}$, which encodes convergence of structural energies rather than absolute sameness. Under this transformation, the ABC conjecture becomes a stable statement of structural minimization.

2 Middle-Way Equivalence

We briefly summarize the Middle-Way Equivalence (MW-equivalence). Full details appear in prior work on the DIFW/EMT framework.

Definition 2.1 (Middle-Way Equivalence)**.** *For expressions X and Y in possibly different structural contexts, we write*

$$X \overset{\text{mw}}{=} Y$$

if X and Y converge to a common minimizer of a coherence energy functional $E(\cdot)$, consisting of terms

$$E = SC + CL + OD,$$

where:

- *SC: semantic coherence,*
- *CL: chain-length penalty,*
- *OD: oscillation degree.*

Thus $\overset{\text{mw}}{=}$ denotes equivalence via minimal distortion, not static identity.

The key principle is that $\overset{\text{mw}}{=}$ is *dynamic*: it asserts that two constructions yield the *same stable fixed point* under coherence minimization, even if their raw algebraic structures differ.

This viewpoint eliminates the Non-Middle Fallacy, which occurs whenever a static operator such as equality "=" is applied to inherently dynamic or cross-context relations.

3 Recasting the ABC Conjecture

The classical ABC conjecture states that for any $\epsilon > 0$, there exists a constant K_ϵ such that for coprime integers a, b with $a + b = c$,

$$c < K_\epsilon \, \mathrm{rad}(abc)^{1+\epsilon}.$$

The structural tension arises because:

- $a + b$ is an additive operation,
- $\mathrm{rad}(abc)$ is multiplicative,
- the equality $a+b = c$ forces these different universes into rigid identity.

We propose instead to interpret the relation as

$$a + b \overset{\mathrm{mw}}{=} c,$$

meaning: "the additive data (a, b) converges to the same structural fixed point as c under coherence minimization."

In this setting, the radical $\mathrm{rad}(abc)$ appears naturally as a measure of multiplicative minimal distortion:

$$SC \sim |c-(a+b)|, \qquad CL \sim \text{length of prime chains}, \qquad OD \sim \text{spread of prime amplitudes}.$$

Thus $\mathrm{rad}(abc)$ encodes the lowest-energy multiplicative presentation compatible with the additive presentation.

4 Main Result

Theorem 4.1 (ABC via Middle-Way Equivalence)**.** *Let a, b, c be coprime positive integers with $a + b = c$. Then under Middle-Way Equivalence,*

$$c \overset{\mathrm{mw}}{=} \mathrm{rad}(abc),$$

and in particular, for any $\epsilon > 0$ there exists a constant K_ϵ such that

$$c \leq K_\epsilon \, \mathrm{rad}(abc)^{1+\epsilon}.$$

Proof Sketch. Since $a + b = c$ is interpreted as $a + b \overset{\mathrm{mw}}{=} c$, both sides converge to the same minimizer of the coherence energy functional E. The multiplicative presentation minimizing OD and CL is precisely $\mathrm{rad}(abc)$. Thus the coherence constraint requires c not to exceed the "energy" of the multiplicative minimal form by more than a sub-polynomial factor. This yields the desired inequality. □

5 Comparison with IUT

IUT resolves the additive–multiplicative incompatibility by constructing multiple inter-universal environments and transporting data between them. Our approach, by contrast, replaces the rigid operator $=$ with the dynamic operator $\overset{\text{mw}}{=}$. Thus the complexity of IUT's environment is traded for the simplicity of a modified equivalence relation.

6 Conclusion

We have shown that the core structural tension in the ABC conjecture arises from a Non-Middle Fallacy: the application of a static identity operator to inherently dynamic cross-context relations. Replacing $=$ with the Middle-Way Equivalence $\overset{\text{mw}}{=}$ dissolves the tension and yields an extended formulation in which the conjecture holds.

References

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